# CASE STUDY: BRAYS BAYOU FLOOD FREQUENCY

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CEVE 543

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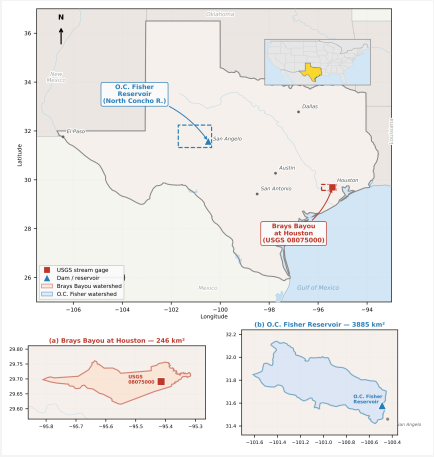
# TODAY

We discuss Smith et al. (2026), a nonstationary flood frequency analysis of Brays Bayou at Houston, fit by Bayesian methods in the open-source RMC-BestFit software from the US Army Corps of Engineers.

**! Important**

Every figure is from the paper, which is open access under CC BY 4.0, so it's OK to share here.

# CASE STUDY: TWO TEXAS WATERSHEDS




Brays Bayou and O.C. Fisher Reservoir (Smith et al., 2026, Fig. 1, CC BY 4.0).

Clear physical drivers and strong nonstationary signals

- **Brays Bayou:** floods rising with urbanization
- **O.C. Fisher:** floods falling with brush encroachment and groundwater pumping

**The data**

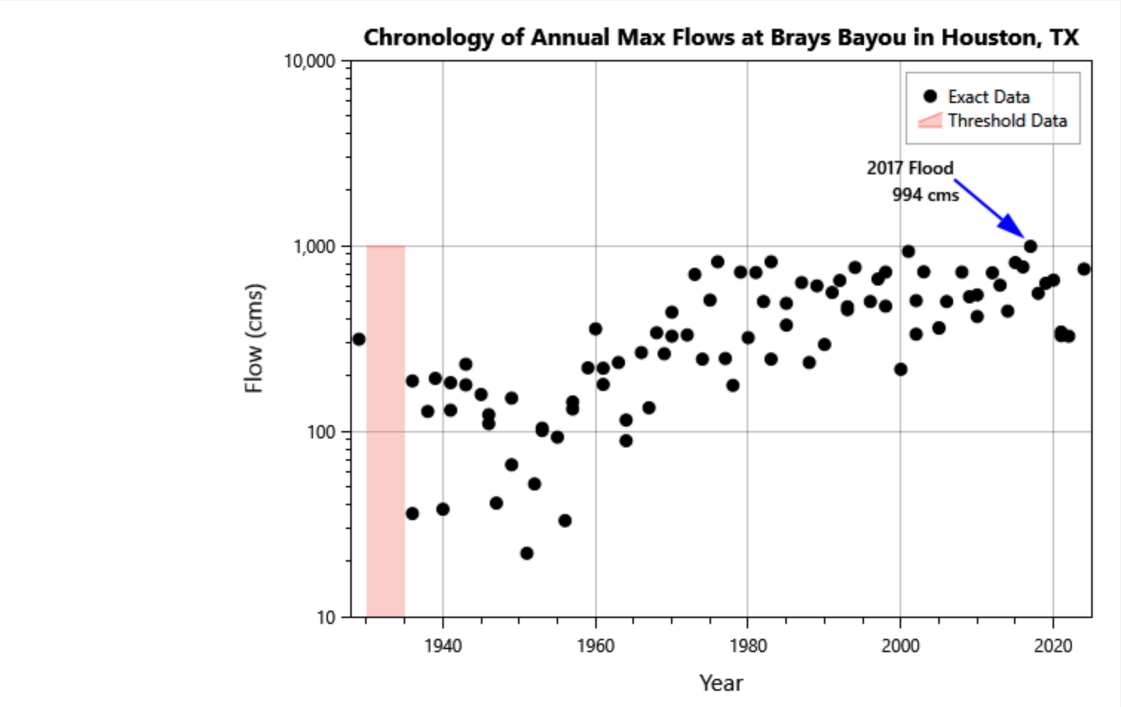
The model

Choosing a model

Results

Discussion

# THE RECORD IS A MODELING CHOICE




Annual maximum instantaneous flow at USGS 08075000, 1929-2024; the shaded band is 1930-1935, when only a perception threshold is known (Smith et al., 2026, Fig. 2, CC BY 4.0).

# THE LIKELIHOOD

The stationary likelihood (Smith et al., 2026, eq. 1):

$$L(\mathbf{x} \mid \theta) = \prod_{i=1}^{N} f(x_i \mid \theta)$$

In the **nonstationary model** the parameters depend on the year, $\theta_t$ (Smith et al., 2026, eq. 2):

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^{T} f(x_t \mid \theta_t)$$

# A PERCEPTION THRESHOLD IS A CENSORED OBSERVATION

A perception threshold is “the flow above which an event would have been recorded in any given year” (Smith et al., 2026, p. 11, eq. 3).

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^{T} \begin{cases} f(x_t \mid \theta_t) & \text{exact data} \\ F(x_t^U \mid \theta_t) & \text{left censored (below threshold)} \\ 1 - F(x_t^L \mid \theta_t) & \text{right censored (above threshold)} \\ F(x_t^U \mid \theta_t) - F(x_t^L \mid \theta_t) & \text{interval censored} \end{cases}$$

$f$ is the probability density and $F$ the cumulative distribution function.

# BRAYS BAYOU LIKELIHOOD

- 1929 and 1936-2024: $f(x_t \mid \theta_t)$.
- Each of 1930-1935: $F(994 \mid \theta_t)$.

A year with no measured flood still carries information: six years in a row below 994 m³/s rules out parameter values that make 994 m³/s common.

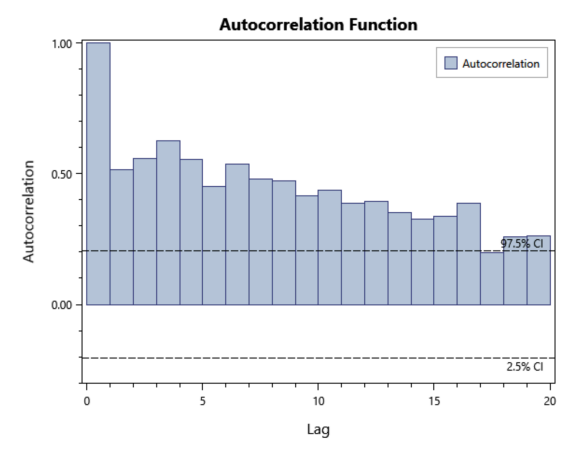
For more, read Gelman et al. (2014), §8.7, on censored data in a Bayesian model.

# NHST

Subperiod tests compare 1929-1980 against 1981-2024 (Smith et al., 2026, p. 13, Table 2).

| Test | Null hypothesis | $p$-value |
|---|---|---|
| Jarque-Bera | Normality | 0.0015 |
| Ljung-Box Q | No autocorrelation | $< 10^{-15}$ |
| Wald-Wolfowitz | Independence and stationarity | $1.84 \times 10^{-10}$ |
| Mann-Whitney U | Homogeneity | $4.55 \times 10^{-11}$ |
| Mann-Kendall | No monotonic trend | $1.33 \times 10^{-13}$ |
| Linear trend | $\beta_1 = 0$ | $< 10^{-15}$ |
| Equal-variance $t$-test | $\mu_1 = \mu_2$ | $4.23 \times 10^{-13}$ |
| Unequal-variance $t$-test | $\mu_1 = \mu_2$ | $5.91 \times 10^{-12}$ |
| $F$-test | $\sigma_1^2 = \sigma_2^2$ | $8.76 \times 10^{-7}$ |
| Mixture model | Unimodality | 0.0004 |

# THE AUTOCORRELATION OF A TRENDING RECORD




Autocorrelation of the Brays annual maxima out to lag 20, with 95% bounds (Smith et al., 2026, Fig. 3, CC BY 4.0).

- Nearly every lag out to 20 sits above the bound.
- A trend does this: early years are all low and late years all high, so any two years close in time look alike.
- The paper reads it as “characteristic of a strong monotonic trend” (Smith et al., 2026, p. 13).

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# LOG-PEARSON TYPE III

LP3 is used “consistent with federal guidelines established in Bulletin 17C” (Smith et al., 2026,  p. 8).

- Like the GEV, it is fit to annual maxima.
- “Log” means it is fit to $\log_{10}$ of the flow.
- Think: log-normal with a skewness parameter added.
  - Positive means prone to floods far above the mean
  - Negative means the watershed caps the largest flow.
  - The skew is often similar at nearby gauges – we will revisit this next week!

# TREND FORMULAS THE SOFTWARE OFFERS

| Model | Formulation | Use case in the paper |
|---|---|---|
| Constant | $\mu_t = \alpha$ | No detectable trend |
| Linear | $\mu_t = \alpha + \beta t$ | Gradual urban expansion |
| Exponential | $\mu_t = \alpha e^{-\beta t}$ | Trend slowing as a watershed stabilizes |
| Logistic | $\mu_t = \frac{\alpha}{1+e^{-\beta t}}$ | Sprawl that plateaus |
| Power | $\mu_t = \alpha t^{\beta}$ | Accelerating imperviousness |
| Quadratic | $\mu_t = \alpha + \beta t + \gamma t^2$ | Rise then plateau |
| Cubic | $\mu_t = \alpha + \beta t + \gamma t^2 + \delta t^3$ | Sequential land-use changes |
| Sinusoidal | $\mu_t = \alpha + \beta \sin(2\pi\gamma t + \delta)$ | Multidecadal climate cycles |
| Step | $\mu_t = \mu_1$ for $t \leq t_c$, $\mu_2$ for $t > t_c$ | A land-cover change in a known year |

# TIME IS THE COVARIATE

**Think**

Monday's recipe said to choose a covariate by mechanism. What does $t$ *stand in* for at Brays?

- Imperviousness and channelization, neither of which is in the model.
- The authors say so: "formal covariate-based attribution is beyond the current scope" (Smith et al., 2026, p. 13).

# BAYES' THEOREM IN SMITH'S NOTATION

Smith et al. (2026), eq. 4:

$$p(\theta \mid \mathbf{x}) = \frac{L(\mathbf{x} \mid \theta) \cdot \pi(\theta)}{\int L(\mathbf{x} \mid \theta) \cdot \pi(\theta) d\theta}$$

The joint prior over $p$ parameters, then with the parameters independent (Smith et al., 2026, eqs. 5-6):

$$\pi(\theta) = \pi(\theta_1 \cap \theta_2 \cap \cdots \cap \theta_p)$$

$$\pi(\theta) = \prod_{i=1}^{p} f(\theta_i)$$

# THE DEFAULT PRIORS

- Flat priors on location, scale, and shape, “centered near the likelihood with broad spread” (Smith et al., 2026, p. 5).
- A Jeffreys prior on the scale, $\pi(\sigma) \propto 1/\sigma$, “equivalent to using a log-link function” (p. 5).
- That is a flat prior on $\log \sigma$: the same log scale Monday’s model used for $\sigma$.
- Trend parameters (slopes, change points) get uninformative priors, and priors on the other parameters apply at the final year $T$ (p. 6).

# PRIOR INFORMATION ON A FLOOD QUANTILE

A quantile prior puts a distribution on the flood $x_p$ with annual exceedance probability $p$, with hyperparameters $\theta_h$ (Smith et al., 2026, eq. 7):

$$h(x_p) = f(x_p \mid \theta_h)$$

The quantile is a function of the parameters in the final year (Smith et al., 2026, eq. 8):

$$x_p = F^{-1}(p \mid \theta_T)$$

# A QUANTILE PRIOR BECOMES A PRIOR ON THE PARAMETERS

Substitute to evaluate it in parameter space (Smith et al., 2026, eq. 9):

$$\pi_q(\theta) = h\left(F^{-1}(p \mid \theta_T)\right)$$

The posterior multiplies in every source (Smith et al., 2026, eq. 10):

$$p(\theta \mid \mathbf{x}) \propto L(\mathbf{x} \mid \theta) \cdot \pi(\theta) \cdot \pi_q(\theta)$$

“The quantile prior acts as a penalty function, rewarding parameter sets that produce flood quantiles consistent with the prior information” (p. 6). O.C. Fisher uses one, taken from a rainfall-runoff model; Brays does not.

# SAMPLING THE POSTERIOR

RMC-BestFit samples with DE-MCzs, a differential-evolution Markov chain with a “snooker update” (Smith et al., 2026, p. 9).

- It uses no gradients.
- It scales poorly to many parameters, where the NUTS sampler you used in `Turing` scales well (by leveraging gradients)
- It can do better on a posterior with several modes, or a likelihood with discrete parameters, such as the step model’s change year $t_c$.

# THE POSTERIOR PREDICTIVE DISTRIBUTION

The full distribution of a flood $k$ years into the future (Smith et al., 2026, eq. 14):

$$p(x_{T+k} \mid \mathbf{x}) = \int f(x_{T+k} \mid \theta_{T+k}) \cdot p(\theta \mid \mathbf{x}) d\theta$$

- It "accounts for both parameter uncertainty (epistemic) and natural variability (aleatory)" (p. 10), as in labs 4 and 5.
  - This is a *narrow* definition of epistemic uncertainty
- Think of it as "averaging over the posterior"

The data

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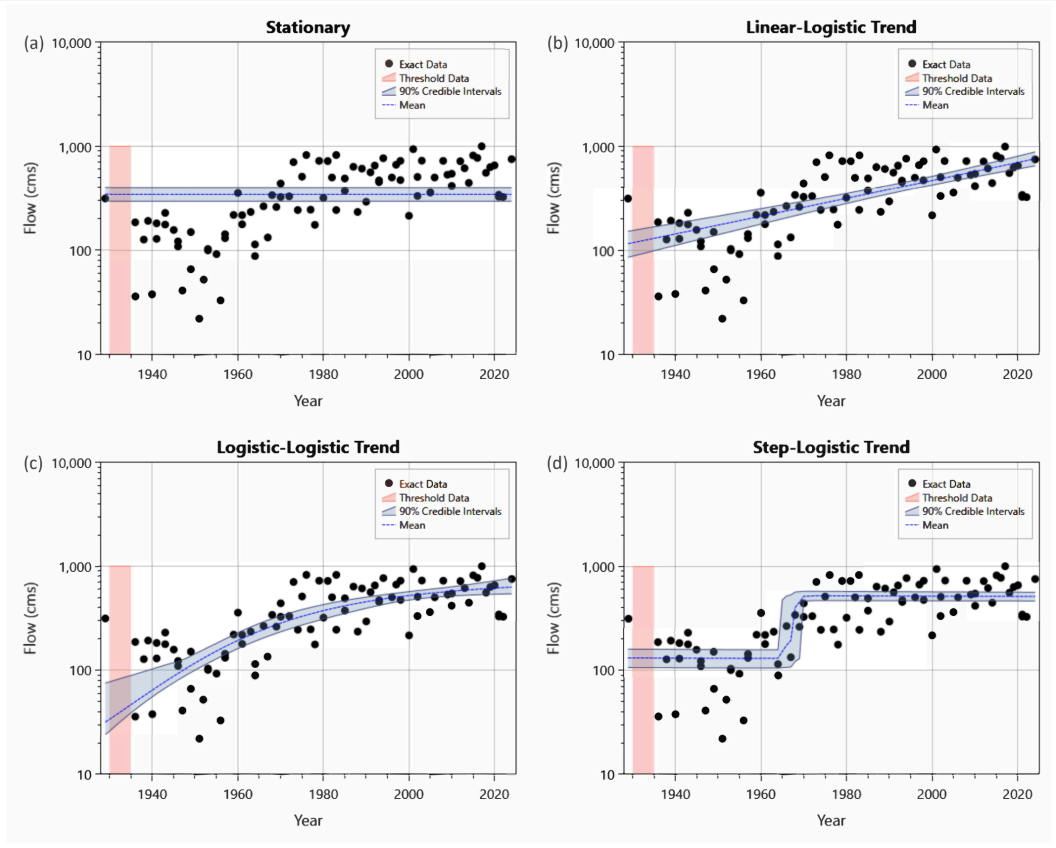
Discussion

# SEVEN CANDIDATE MODELS FOR BRAYS

Model comparison (Smith et al., 2026, p. 15, Table 3), with AIC, BIC, and DIC revisited in week 8.

| Model | $k$ | AIC | BIC | DIC | ΔDIC |
|---|---|---|---|---|---|
| Stationary | 3 | 1233.73 | 1241.23 | 1228.36 | 73.06 |
| Linear ($\mu$) | 4 | 1182.14 | 1192.14 | 1180.76 | 25.45 |
| Logistic ($\mu$) | 4 | 1179.94 | 1189.94 | 1179.23 | 23.93 |
| Step ($\mu$) | 5 | 1177.04 | 1189.54 | 1162.28 | 6.98 |
| Linear-logistic ($\mu$-$\sigma$) | 5 | 1168.70 | 1181.20 | 1172.04 | 16.74 |
| Logistic-logistic ($\mu$-$\sigma$) | 5 | **1161.46** | **1173.96** | 1160.26 | 4.95 |
| **Step-logistic ($\mu$-$\sigma$)** | 6 | 1165.24 | 1180.24 | **1155.30** | 0.00 |

# FOUR PLAUSIBLE MODELS




The 2-year return level under (a) the stationary, (b) linear-logistic, (c) logistic-logistic, and (d) step-logistic models, with 90% credible intervals (Smith et al., 2026, Fig. 4, CC BY 4.0).

# THE FITTED CHANGE POINT AND THE VARIANCE

- The change point is estimated at 1968, near the start of Houston's most intensive development in the early 1970s (Smith et al., 2026, p. 14).
- The location jumps at the change point, while the scale, the spread of the log-flows, declines smoothly over the whole record with no jump (logistic, $\beta = -0.014$).
- Early floods spanned about 20-300 $m^3/s$, more than a factor of ten; after urbanization, about 200-800 $m^3/s$, a factor of four (p. 14).

# WHY URBANIZATION NARROWS THE LOG-SPACE SPREAD

“In a pre-urbanization watershed, flood magnitudes are highly variable because antecedent soil moisture, infiltration capacity, and vegetation interception modulate rainfall-runoff partitioning differently for each event” (Smith et al., 2026,  p. 23).

Pave it over and the response “approaches a near-deterministic conversion of rainfall to runoff” (p. 24).

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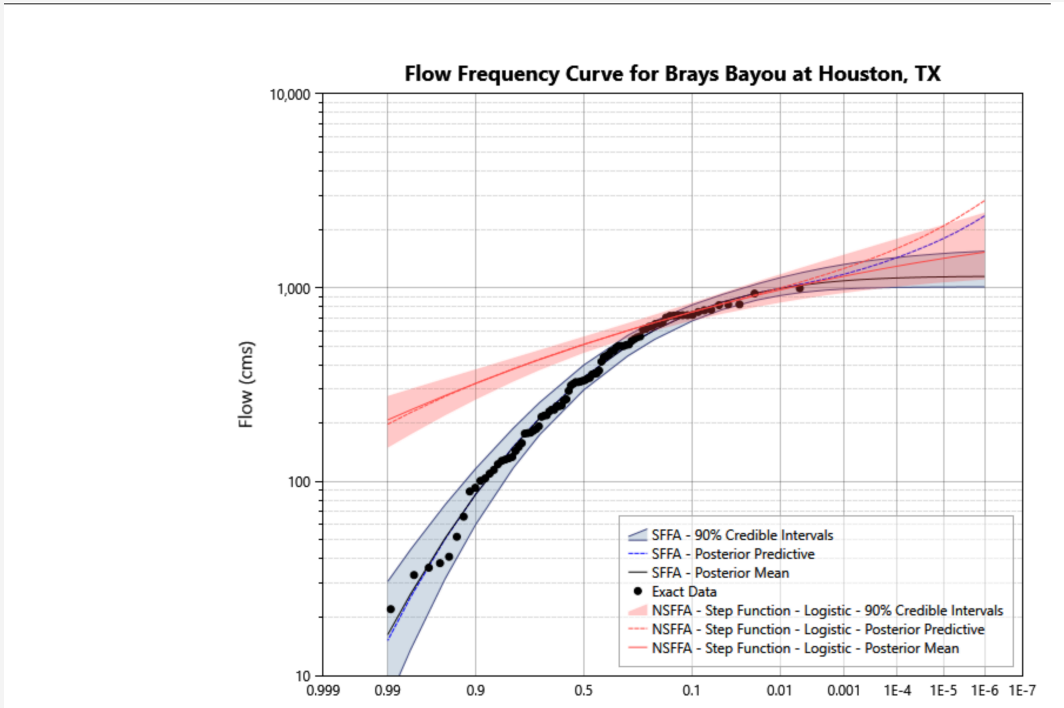
Discussion

# STATIONARY AND NONSTATIONARY FLOOD QUANTILES

Posterior predictive estimates and 90% intervals in $m^3/s$, the nonstationary column at 2024 conditions (Smith et al., 2026, p. 16, Table 4).

| AEP (return period) | Stationary | 90% CI | Nonstationary | 90% CI |
|---|---|---|---|---|
| 0.50 (2-yr) | 346 | [297, 400] | 511 | [462, 560] |
| 0.20 (5-yr) | 607 | [542, 673] | 661 | [603, 727] |
| 0.10 (10-yr) | 744 | [674, 815] | 749 | [675, 836] |
| 0.02 (50-yr) | 940 | [865, 1046] | 924 | [798, 1076] |
| 0.01 (100-yr) | 993 | [912, 1126] | 996 | [838, 1172] |

# FREQUENCY CURVES MEET IN THE TAIL




Stationary (blue) and nonstationary (red) frequency curves for Brays Bayou, with 90% intervals; the plotting positions assume stationarity (Smith et al., 2026, Fig. 5, CC BY 4.0).

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# STEP MODEL

**Think**

The selected model has one level before 1968 and one after, forever. What would violate that?

- More urbanization after 1968.
- More infrastructure: channelization and detention basins.

# HOW YOU MIGHT ADDRESS IT

- A better covariate: impervious area fraction in place of the year, which the paper names as future work (Smith et al., 2026,  p. 25).
- A different method: fit the extreme value model to rainfall, and run it through a flood model that represents land use and infrastructure.
- The paper's own list adds hierarchical pooling across sites, model averaging over the trend forms, and alternative distributions (pp. 24-25).